



Laboratory and field evaluation of an autoinoculation device as a tool to manage poultry red mite, *Dermanyssus gallinae*, infestations with *Beauveria bassiana*

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Abstract

The poultry red mite, *Dermanyssus gallinae*, is a cosmopolitan ectoparasite in hens and has been considered an important threat to the egg production industry. This study evaluated an alternative to manage poultry red mite populations as a complement to conventional chemical treatments and other control strategies in poultry houses. A simple autoinoculation device prepared with corrugated cardboard (CB) or loofah sponge (LS) as inert supports to anchor *Beauveria bassiana* conidia was used to aggregate and infect mites from infested poultry houses. In the laboratory, mites gathered inside the traps and the average mortalities by the fungus were higher than 70% in CB and LS traps after 5 and 4 days of exposure, respectively. Conidial viability was around 80% in CB and LS traps after 14 and 60 days under unrefrigerated conditions (26 °C), respectively. Both trap types tied to hen cages efficiently captured fed mites after blood meal, and fungal infection was observed in 65–90% of the mites in field tests. Between 5 and 25% of the mites recaptured in monitoring cardboard traps installed immediately after CB and LS removal were infected by *B. bassiana*. According to our results, the use of *B. bassiana* in an autoinoculation strategy is a potential alternative method for *D. gallinae* control.

Keywords Microbial control · Attract-and-infect · Poultry house · Animal production

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Introduction

The poultry red mite, *Dermanyssus gallinae* (De Geer), is considered one of the most important ectoparasites of laying hens worldwide (Chauve 1998; Sparagano et al. 2014; George et al. 2015; Pritchard et al. 2015), and the impact of infestation by this blood-feeding mite significantly affects the productivity of the egg industry and poses serious animal health and welfare concerns (Kilpinen et al. 2005; Flochlay et al. 2017). Chemical acaricides are broadly used to mitigate the poultry red mite infestation, but the limited number of available products and the widespread mite resistance to these acaricides have increased the prevalence rates of this pest (Abbas et al. 2014; Sparagano et al. 2014). Moreover, the misuse of non-registered products has been responsible for animal, egg, and environment contaminations, increasing the risks to human health (Sparagano et al. 2009; Flochlay et al. 2017). Residues of several pesticides have already been found worldwide contaminating organs and tissues of laying hens as well as in eggs (Hamscher et al. 2003; Tao et al. 2009; Marangi et al. 2012; Eladl et al. 2017).

The aggregation behavior of *D. gallinae* in cracks and crevices may also affect the efficacy of synthetic acaricides used in conventional spraying. This behavior is mediated by chemical volatile cues (aggregation pheromones), when mixed developmental stages of the mite return to their hiding places after blood meal during the night (Entrekin and Oliver 1982; Koenraad and Dicke 2010). However, the gregarious and concealed lifestyle of *D. gallinae* may also be exploited to develop attract-and-kill strategies. Cardboard traps, first suggested for monitoring population dynamics of the red mite in poultry houses (Nordenfors and Chirico 2001; Lammers et al. 2017), has been used with different killing agents, such as chemical acaricides (Chirico and Tauson 2002) and plant extracts (Lundh et al. 2005; Barimani et al. 2016).

Acaropathogenic fungi are able to infect and kill various mite species, including blood-feeding mites (Chandler et al. 2000; Fernandes et al. 2012; Beys-da-Silva et al. 2020). Some fungal species have been proven effective against *D. gallinae* under lab and field conditions by directly spraying conidial suspensions (Kaoud 2010; Tavassoli et al. 2011; Immediato et al. 2015, 2017; Kasburg et al. 2016) or in association with inert dusts (Steenberg and Kilpinen 2014) and essential oils (Immediato et al. 2016). However, these microbial agents have never been evaluated in association with traps in an attract-and-infect strategy against the red mite. The environmentally friendly characteristics of biological control agents make them candidates to incorporate into this strategy. Therefore, the aim of our study was to develop two types of traps containing *Beauveria bassiana* conidia and to test the efficacy of these traps to attract and infect *D. gallinae* under laboratory conditions and in mite-infested poultry houses.

Materials and methods

Mites and fungal strain

Mites were collected from an infested poultry house, which had not been chemically treated, located in southern Brazil (Paraná State, S 25° 26' 08.4", W 54° 04' 39.9"). Blood-fed mites were transferred to plastic bags using a brush, and only adult females showing intense red color were selected for the lab bioassays.

The strain Unioeste 88 of *B. bassiana* was identified by the analysis of the Bloc nuclear intergenic region sequence (Rehner et al. 2011). It was originally isolated from an infected Cerambycidae beetle and was previously selected as virulent to *D. gallinae* (Kasburg 2016). The fungus was grown on potato dextrose agar medium (PDA) for 7–10 days at 26 ± 0.5 °C and L12: D12 photoperiod. Conidia were scraped with a spatula and used as inoculums in liquid and solid fermentation for traps preparation.

Preparation of traps

Two autoinoculation devices containing different inert supports to anchor *B. bassiana* conidia were prepared for the laboratory and field experiments. In the first case, conidia were produced by solid-state fermentation on autoclaved parboiled rice loaded into polypropylene bags. Briefly, rice was soaked in warm distilled water (ca. 1:1 water–rice w/w) for 10 min and then autoclaved for 20 min in polyethylene bags (200 g rice per bag). Each bag was inoculated with 10 mL of a conidial suspension from PDA medium [ca. 1×10^7 conidia per mL of water + the surfactant Tween 80 (0.05% v/v)]. Rice in bags was agitated and bags were incubated for 10–12 days at 26 ± 2 °C and L12:D12 photoperiod. After this period, bags were opened and the substrate was allowed to air dry for 48 h, after which conidia were extracted by sieving the fungus-colonized rice. Harvested conidia were dried by exposure to desiccant for 7 days at 25 °C in hermetic containers (ca. 3 g of conidia in a 600-mL container filled with 300 g of silica gel). Samples of dry conidia (0.01 g) were suspended in 9.99 mL of water + Tween 80 (0.05% v/v) and counted using a Neubauer chamber after serial dilution to assess the concentration of conidia in the dry powder. For conidial viability assessment, a 50- μ L drop of a suspension containing ca. 1×10^5 conidia mL⁻¹ was placed on PDA medium and plates were incubated at 25 ± 0.5 °C in L12: D12 photoperiod. Germination was assessed by direct microscopic observation ($\times 400$ magnification) of 300 conidia after 18–20 h of incubation. Conidia were considered germinated if a germ tube larger than conidial length was visible. Dry powder had ca. 5×10^{10} conidia g⁻¹ and conidial viabilities were above 90% at the time they were used. Corrugated cardboard (CB) pieces (5 $\times$ 6 cm) were treated or not with a thin layer (ca. 0.05 g) of emulsified wax (SPLAT™ - ISCA Tecnologias, Brazil) on one of the cardboard sides, spreading the material homogeneously with a spatula. The wax was used to increase the amount of conidia adhered to the CB pieces. Groups of five CB pieces were placed inside a glass jar containing 1 g of the conidial powder and shaken for 2 min. The amount of powder adhered to the paper and the number of conidia per trap was estimated by weighting the CB piece before and after exposure to conidial powder using a digital analytical balance with 0.0001 g readability. The final trap was prepared rolling and placing the treated inert supports inside PVC tubes (1.6 cm diameter, 5 cm long) (Lammers et al. 2017).

In the second case, loofah sponge (LS) *Luffa aegyptiaca* Mill. (Cucurbitaceae) was used as inert support to anchor fungal cells produced by liquid-state fermentation. Erlenmeyer flasks containing 150 mL of autoclaved liquid medium (41.7 g yeast extract, 52 g sucrose, 1 g KCl, 0.36 g KH₂PO₄, and 0.6 g MgSO₄ per L of distilled water) were inoculated with *B. bassiana* conidia (0.01 g) and incubated for 4 days in an orbital shaker (200 rpm and 25 ± 0.5 °C). Autoclaved cylindrical segments of LS (1 cm diameter, 5 cm long) were immersed in the fermented broth and then held in a vertical position for 10 s to remove the excess liquid. Segments were then placed individually inside 50 mL-Falcon tubes and incubated at 26 ± 2 °C and L12:D12 photoperiod for 5 more days for conidia to form on LS surface. Gas exchange was provided by opening the tube once a day for 2 min in a sterile flow chamber. After this period, tubes were

opened and dried by exposure to desiccant (silica gel) for 7 days at 20 °C in hermetic containers. Traps were assembled also placing the treated LS inside the PVC tubes. The number of conidia per LS was estimated by washing the inert support in 10 mL of distilled water + surfactant (Tween 80, 0.01%) and counting the cells in a Neubauer chamber.

Table 1 illustrates the set of laboratory and field experiments performed to evaluate the efficacy of the autoinoculation strategy using the fungus *B. bassiana* against the poultry red mite *D. gallinae*.

Efficacy of *Beauveria bassiana* traps to infect *Dermanyssus gallinae* in the laboratory

Bioassays with CB and LS traps were conducted separately. First, the infectivity of fungal treated CB traps with (CB-FW) or without (CB-F) wax was compared and untreated CB traps with (CB-CW) or without (CB-C) wax were used as negative controls. In this case, fungal-treated traps were individually placed in glass jars and kept at 26 and 32 ± 2 °C, 60% RH for 7, 14 and 21 days, and after each period, traps were tested against the mites. The higher temperature was selected to simulate poultry house conditions. Next, fungal impregnated LS traps (LS-F) were compared to CB-F traps immediately after preparation for their ability to cause mite infection, and CB-C and LS traps without conidia (LS-C) were used as negative controls.

For both cases, traps were then individually placed inside glass tubes (2.5 cm diameter, 9 cm long) with 30 engorged females and closed with a voile cloth (4×4 cm) (Kasburg et al. 2016). Tubes were kept in a vertical position under controlled conditions in an incubator (26 ± 2 °C, L12:D12 photoperiod, and ca. 60% RH). Four replicates were prepared for each treatment. The number of dead mites (individuals with no movement when touched) was evaluated daily up to 4 and 6 days for bioassays with LS and CB traps, respectively. Cadavers were collected on the last day of evaluation and washed in an ethanol solution (70%) for 5 s, rinsed in distilled water and kept in a moistened chamber. After 3–4 days cadavers were examined macro and microscopically to confirm the fungal infection. Both bioassays were performed twice on different dates and with independent trap batches.

Stability of *Beauveria bassiana* conidia in the autoinoculation devices under unrefrigerated conditions

The viability of the conidia in both autoinoculation devices was evaluated over time under the unrefrigerated conditions commonly found in poultry houses. Traps were individually placed in glass jars and exposed to 26 ± 2 °C, 60% RH for 1, 7, 14, 35, 60 and 90 days, and to 26 and 32 ± 2 °C, 60% RH for 7, 14 and 21 days for LS and CB traps, respectively. Four samples were prepared for each time and trap type. Conidia were washed from traps in distilled water + surfactant (Tween 80, 0.01%) and a 50-μL aliquot of the suspension inoculated onto PDA medium. Petri dishes were then incubated at 25 ± 0.5 °C for 18–20 h and conidial germination was assessed as already described.

Efficacy of *Beauveria bassiana* traps to infect *Dermanyssus gallinae* populations in poultry houses

Experiments were performed in infested poultry houses at two locations. The first experiment (field trials 1 and 2) was conducted in Paraná State (S 25° 10' 35.0", W 53° 53' 25.2") in November 2018 in a commercial poultry house with metal cages on a wood made structure

Table 1 Laboratory and field experiments testing autoinoculation traps impregnated with the fungus *Beauveria bassiana* against the poultry red mite, *Dermanyssus gallinae*

Experiment	Trap	Experimental conditions and parameters
1. Efficacy of cardboard traps (CB) impregnated with <i>B. bassiana</i> as an autoinoculation device against <i>D. gallinae</i> in the lab	CB with or without wax and impregnated with dry conidia	<i>Conditions</i> traps stored for 7, 14 and 21 days at $26 \pm 2^\circ\text{C}$ <i>Parameters</i> mortality of engorged females and conidial viability after each storage period <i>Conditions</i> traps stored for 7, 14 and 21 days at $32 \pm 2^\circ\text{C}$ <i>Parameters</i> mortality of engorged females and conidial viability after each storage period
2. Efficacy of two autoinoculation devices against <i>D. gallinae</i> in the lab	Loofah sponge (LS) and CB traps impregnated with fungal conidia	<i>Conditions</i> CB and LS traps incubated at $26 \pm 2^\circ\text{C}$ <i>Parameters</i> mortality of engorged females <i>Conditions</i> LS traps stored for 90 days at $26 \pm 2^\circ\text{C}$ <i>Parameters</i> conidial viability
3. Efficacy of two autoinoculation devices against <i>D. gallinae</i> populations in poultry houses	LS and CB traps impregnated with fungal conidia	<i>Conditions</i> CB traps tied to hen cages <i>Parameters</i> mortality of mites collected after 12, 24 and 48 h <i>Conditions</i> LS traps tied to hen cages <i>Parameters</i> mortality of mites collected after 12, 24 and 48 h

(4–6 hens per cage and a total of 15 thousand 60-week-old hens) and had not been chemically treated. Three groups of 10 LS-F traps (capture trap) were randomly tied to the cages throughout the house (only one trap per cage) and not less than 3 m between traps, always in cages next to mite agglomerations and just before sunset (ca. 18:00 h). Groups of LS-C traps were used as negative control treatment. Each group of LS-F and LS-C represented the intervals of time that traps were collected. After 12, 24 and 120 h one group of LS-F and one group of LS-C traps were collected and replaced by an untreated CB trap (recapture trap), which remained for an additional 24 h to evaluate the presence of infected individuals in mite population after treatment. Temperature and relative humidity of the poultry houses were monitored during the experimental period 3×per day (08:00, 13:00 and 18:00 h) using a thermohydrometer (average 27 ± 4 °C, $68 \pm 15\%$ RH). In the laboratory, 30 engorged females were collected from each of the collected traps, transferred to glass tubes and kept at 26 ± 2 °C, 60% RH, and L12:D12 photoperiod for 7 days. The number of dead mites was evaluated daily up to 6 days as already described. The whole experiment was performed twice on different dates and with independent trap batches.

The second experiment (field trials 3 and 4) was conducted in Paraná State (S 25° 15' 08.9", W 53° 56' 32.8") in December 2018 and January 2019 in a commercial poultry house with metal cages (4–6 hens per cage and a total of 20 thousand 72-week-old hens) and had not been chemically treated. In this case, three groups of 15 CB-FW traps (capture trap) were also randomly tied to the cages (ca. 17:00–18:00 h). Groups of CB-C traps were used as negative control treatment. Each group of CB-FW and CB-C represented the intervals of time that traps were collected. After 12, 24, and 48 h, one group of each trap type was collected and replaced by an untreated CB trap (recapture trap), which remained for an additional 24 h. Temperature and relative humidity of the poultry houses were monitored during the entire period using a thermohydrometer (30 ± 8 °C, $68 \pm 15\%$ RH). Trap evaluation in the laboratory followed the same procedure described for the first field trial. The whole experiment was performed twice on different dates and with independent trap batches.

Statistical analysis

Conidial stability under storage and mite mortality experiments in the laboratory were arranged in a completely randomized design, and analyses were performed using Statsoft software. The assumption of normality was checked for all the experiments (Shapiro–Wilk), and conidial viability and mite mortality data over time were analyzed by a repeated measures ANOVA mixed model, where treatments were considered as fixed effect in the model and time as random effect. Mean values were statistically separated by Tukey's honestly significant difference (HSD) test (or N-HSD if N was different) at $\alpha=0.05$. For the field trials, experiments were arranged in a randomized block design and mite mortality data were also analyzed by a repeated measure ANOVA, separately for capture and recapture data. Mean values were statistically separated by Tukey's HSD test (or N-HSD).

Results

Efficacy of *Beauveria bassiana* traps to infect the poultry red mite *Dermanyssus gallinae* in the laboratory

Anchorage of vegetative cells from fermented broth on LS was homogeneous and conidiation occurred all over the web structure of the support, with an average of $2.65 \pm 0.25 \times 10^9$ conidia per trap. For CB traps, averages of $1.06 \pm 0.03 \times 10^9$ and $0.509 \pm 0.02 \times 10^9$ conidia per trap were adhered to the support either treated or not with paraffin wax, respectively, after exposure to conidial powder.

Significant interaction was seen between cardboard traps and storage time on mite mortality at 26 °C ($F_{3,84} = 14.99$, $p < 0.0001$). Average mite mortalities were higher in CB-F traps compared to the control groups and fungal activity remained high (> 80%) up to 14 days. Differences were also detected for confirmed mite mortality among treatments ($F_{3,84} = 20.78$, $p < 0.0001$) (Table 2). The same trend was seen at 32 °C, and significant interactions between cardboard traps and storage time on total ($F_{3,81} = 18.02$, $p < 0.0001$) and confirmed ($F_{3,84} = 20.10$, $p < 0.0001$) mite mortality were observed. Mite mortalities in CB-F traps were different from control traps, and the fungal activity was affected by the storage time, maintaining high (> 80%) only up to 7 days (Table 3). Cardboard traps treated with wax and fungal conidia did not differ from traps treated only with dry conidia at both temperatures and most dead mites in CB-F and CB-FW treatments showed signs of fungal infection (mummification) (Tables 2 and 3). Overall mite mortality was significantly higher in experiments conducted at 26 °C than at 32 °C ($F_{1,55} = 8.54$, $p = 0.0050$) regardless of CB trap type (treated or not with the fungus and/or the wax), but differences were not detected when CB trap types were considered ($F_{3,165} = 0.24$, $p = 0.87$).

Significant differences were detected among fungal treated and untreated traps on total ($F_{1,26} = 14.73$, $p = 0.0007$) and confirmed ($F_{1,26} = 17.33$, $p = 0.0003$) mite mortality. LS-F

Table 2 Effect of storage time at 26 °C of *Beauveria bassiana*-treated cardboard traps (CB) on subsequent mean ($\pm$ SD) cumulative mortality (%) of *Dermanyssus gallinae* in the laboratory (assessed daily for up to 6 days post-exposure)

	Treatment	N	Days under storage at 26 °C			
			0	7	14	21
Total mortality (%) ^a	CB-C	8	5.00 $\pm$ 3.09bB	10.83 $\pm$ 2.36bB	7.08 $\pm$ 5.76bB	7.50 $\pm$ 4.27bB
	CB-CW	8	5.83 $\pm$ 3.45bB	8.33 $\pm$ 5.35bB	7.50 $\pm$ 4.63bB	7.92 $\pm$ 6.65bB
	CB-F	8	94.90 $\pm$ 3.17aA	96.22 $\pm$ 5.49aA	82.16 $\pm$ 11.96aB	75.20 $\pm$ 10.81aB
	CB-FW	8	89.03 $\pm$ 6.21aA	93.46 $\pm$ 7.76aA	86.56 $\pm$ 9.27aA	75.06 $\pm$ 5.54aB
Confirmed mortality (%) ^a	CB-C	8	0.0bA	0.0bA	0.0bA	0.0bA
	CB-CW	8	0.0bA	0.0bA	0.0bA	0.0bA
	CB-F	8	92.76 $\pm$ 3.86aA	94.14 $\pm$ 7.51aA	76.84 $\pm$ 8.18aB	70.72 $\pm$ 10.94aB
	CB-FW	8	86.43 $\pm$ 5.86aA	89.90 $\pm$ 10.74aA	83.96 $\pm$ 8.63aA	72.02 $\pm$ 5.13aB

Means followed by different letters in a column (lowercase) or in a row (uppercase) are significantly different (Tukey's HSD: $p < 0.05$)

C control, W wax, F fungus

^aTotal mortality—dead mites (no movement) regardless the presence of fungal structures on the cadavers; Confirmed mortality—assessed using ethanol-washed mites held for fungal emergence and confirmation

Table 3 Effect of storage time at 32 °C of *Beauveria bassiana*-treated cardboard traps (CB) on subsequent mean ($\pm$ SD) cumulative mortality (%) of *Dermanyssus gallinae* in the laboratory (assessed daily for up to 6 days post-exposure)

	Treatment	N	Days under storage at 32 °C			
			0	7	14	21
Total mortality (%) ^a	CB-C	8	5.00 $\pm$ 3.09bA	6.67 $\pm$ 4.36bA	5.00 $\pm$ 3.09bA	6.67 $\pm$ 7.13bA
	CB-CW	8	5.83 $\pm$ 3.45bA	5.83 $\pm$ 4.63bA	6.67 $\pm$ 5.91bA	6.67 $\pm$ 6.17bA
	CB-F	8	94.90 $\pm$ 3.17aA	93.43 $\pm$ 5.14aA	77.09 $\pm$ 9.97aB	75.18 $\pm$ 11.29aB
	CB-FW	7	89.00 $\pm$ 6.71aB	89.18 $\pm$ 5.91aB	74.81 $\pm$ 12.14aB	70.14 $\pm$ 8.66aB
Confirmed mortality (%) ^a	CB-C	8	0.0bA	0.0bA	0.0bA	0.0bA
	CB-CW	8	0.0bA	0.0bA	0.0bA	0.0bA
	CB-F	8	92.76 $\pm$ 3.86aA	91.65 $\pm$ 5.77aA	75.81 $\pm$ 10.81aB	73.50 $\pm$ 8.99aB
	CB-FW	7	86.02 $\pm$ 6.21aA	86.06 $\pm$ 7.22aA	72.86 $\pm$ 11.93aB	69.19 $\pm$ 8.40aB

Means followed by different letters in a column (lowercase) or in a row (uppercase) are significantly different (Tukey's HSD: $p < 0.05$)

C control, W wax, F fungus

^aTotal mortality—dead mites (no movement) regardless the presence of fungal structures on the cadavers; Confirmed mortality—assessed using ethanol-washed mites held for fungal emergence and confirmation

traps caused higher mite mortality than CB-F traps, killing almost 80% of the individuals after 4 days of exposure (Table 4).

Stability of *Beauveria bassiana* conidia in the autoinoculation devices under unrefrigerated conditions

No differences in conidial viability were detected among cardboard trap types kept at 26 vs. 32 °C ($F_{3,84} = 1.53$, $p = 0.21$). However, the storage time influenced the percentage of germinated conidia ($F_{3,84} = 4.00$, $p = 0.010$) for CB-F and CB-FW at both temperatures. The increase in the storage time reduced the conidial viability after 7 days at 32 °C

Table 4 Mean ($\pm$ SD) total and confirmed mortalities (%) of *Dermanyssus gallinae* caused by *Beauveria bassiana* in cardboard (CB) and loofah sponge (LS) traps at 26 °C in the laboratory (assessed daily for up to 6 days post-exposure)

Treatment	N	Total mortality (%) ¹	Confirmed mortality (%) ^a
CB-C	8	13.33 $\pm$ 7.13c	0.0c
LS-C	9	11.11 $\pm$ 9.57c	0.0c
CB-F	4	50.00 $\pm$ 13.61b	43.33 $\pm$ 14.14b
LS-F	9	79.26 $\pm$ 12.67a	72.59 $\pm$ 13.92a

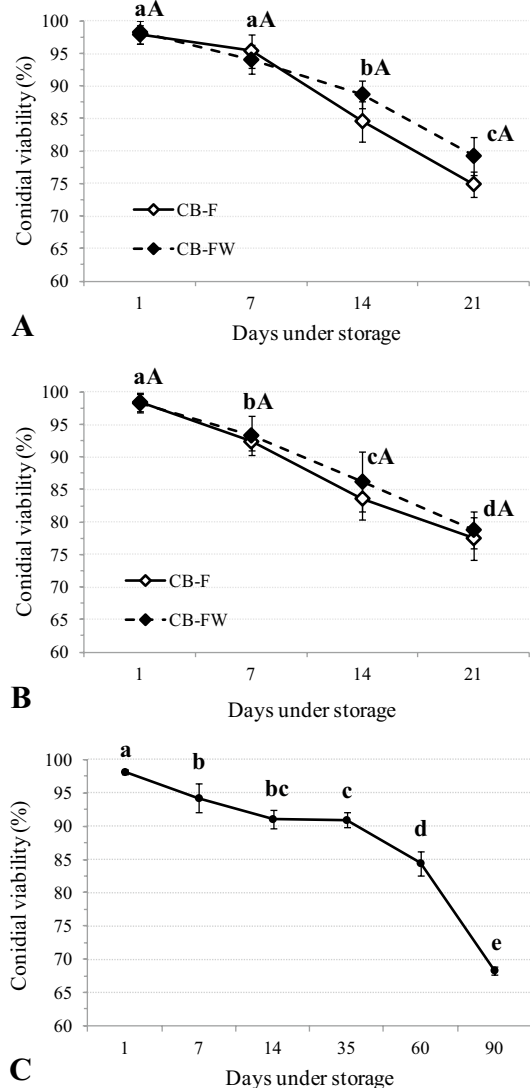
Means followed by different letters in a column (lowercase) or in a row (uppercase) are significantly different (Tukey's HSD: $p < 0.05$)

C control, F fungus

^aTotal mortality—dead mites (no movement) regardless the presence of fungal structures on the cadavers; Confirmed mortality—assessed using ethanol-washed mites held for fungal emergence and confirmation

and after 14 days at 26 °C in both CB-F and CB-FW. After 3 weeks, viability dropped from 98% to approximately 77% (21% of viability loss) in all traps and temperatures (Fig. 1a, b). Decreased conidial viability was also seen in loofah sponge traps maintained at 32 °C ($F_{5,15} = 234.09$, $p < 0.0001$). Loss in viability ranged from 7 to 30% for 14 and 90 days, respectively (Fig. 1c).

Fig. 1 Mean ($\pm$ SD) viability (%) of *Beauveria bassiana* conidia in cardboard traps with (CB-FW) or without (CB-F) wax kept at 26 °C (a) and 32 °C (b), and in loofah sponge trap kept at 26 °C (c). Means capped with different letters are significantly different between storage time (lowercase) or CB trap type (uppercase) (Tukey's HSD: $p < 0.05$)



Efficacy of *Beauveria bassiana* traps to infect *Dermanyssus gallinae* populations in poultry houses

In experiments with LS traps, differences on total mortality ($F_{1,15}=160.51$, $p<0.0001$) and confirmed mortality ($F_{1,15}=180.64$, $p<0.0001$) were observed between fungal treated and untreated capture traps in field trial 1. Mite mortality was significantly higher in LS-F than LS-C in all evaluations, reaching 66% in LS-F traps, although differences on mortality among times after traps installation were not detected. Total ($F_{1,15}=5.71$, $p=0.03$) and confirmed ($F_{1,15}=31.55$, $p<0.0001$) mortalities were different in recapture traps between treated and untreated cages and with LS traps. Between 20 and 23% of the mites collected in recapture traps were found to have been killed by the fungus. Differences in confirmed mortality among evaluation times were also not detected (Table 5). The same trend was observed in field trial 2, where differences were in total mite mortality ($F_{2,34}=3.56$, $p=0.039$) and confirmed mortality ($F_{2,34}=5.50$, $p=0.009$) between LS-F and LS-C capture traps and among times after traps installation. In this case, the number of dead mites decreased in traps maintained longer in the field. Recapture traps also indicated differences in total ($F_{2,34}=8.16$, $p=0.001$) and confirmed ($F_{2,34}=19.12$, $p<0.0001$) mite mortalities in both trap type and trap age, as seen for capture traps. Infected mites in recapture traps reached 25.2 and 4.8% after 1 and 5 days of traps installation (Table 5).

Total mortality ($F_{2,72}=6.81$, $p=0.0019$) and confirmed mortality ($F_{2,72}=3.81$, $p=0.027$) differed between fungal treated and untreated capture CB traps and age of traps in field trial 3. Recapture traps also showed differences in total ($F_{1,74}=20.69$, $p<0.0001$) and confirmed ($F_{1,74}=46.64$, $p<0.0001$) mite mortalities for CB-C and CB-F traps. Mite mortality in CB-F recapture traps was higher than CB-C 12 h after installation of traps, but similar after 24 and 48 h (Table 6). In field trial 4, differences on total mite mortality ($F_{2,114}=6.23$, $p=0.0027$) and confirmed mortality ($F_{2,114}=4.43$, $p=0.014$) between CB-F and CB-C capture traps and among times after traps installation were also observed. The same differences were detected in recapture traps for total ($F_{2,114}=3.014$, $p=0.05$) and confirmed ($F_{2,144}=6.86$, $p=0.0015$) mortalities. As in the experiments with LS traps, a decrease in the total number of dead mites in CB-F traps kept longer was observed in both field trials (Table 6).

Discussion

We evaluated a new strategy to reduce *D. gallinae* populations in infested poultry houses combining a highly virulent strain of the fungus *B. bassiana* with a simple device used to attract and infect mites. Both cardboards impregnated with dry conidia and loofah sponge used as inert support to fungal growth and reproduction effectively attracted and killed the mites by fungal infection under laboratory and field conditions. This is the first report of the use of a biological trap to control poultry red mite populations in caged laying hens.

The attract-and-infect strategy against the gregarious *D. gallinae* is viable using an autoinoculation device impregnated with a biocontrol agent, as already suggested for other chemical and natural products with significant reductions of mite populations in the field after continuous applications (Chirico and Tauson 2002; Lundh et al. 2005; Barimani et al. 2016). We did not assess in our field studies the fluctuation of mite populations in the poultry houses; however, CB and LS traps proved efficient to aggregate

Table 5 Mean ($\pm$ SD) total and confirmed mortalities (%) of *Dermanyssus gallinae* caused by *Beauveria bassiana* in loofah sponge (LS) traps tied to hen cages in a poultry house (assessed daily for up to 6 days post-exposure)

Trap type		N	Days			
			12 h	24 h	5 days	
Field trial 1	Capture trap (LS)	LS-C	9	2.99 $\pm$ 2.61bA	3.33 $\pm$ 7.07bA	-
		LS-F	8	66.77 $\pm$ 28.53aA	70.83 $\pm$ 20.61aA	-
		LS-C	9	0.0bA	0.0bA	-
	Recapture trap (CB)	LS-F	8	63.02 $\pm$ 27.15aA	66.25 $\pm$ 19.96aA	-
		LS-C	9	6.57 $\pm$ 5.14aA	16.30 $\pm$ 19.54aA	-
		LS-F	8	22.50 $\pm$ 10.80aA	27.92 $\pm$ 17.54aA	-
Field trial 2	Capture trap (LS)	LS-C	9	0.0bA	0.0bA	-
		LS-F	8	20.42 $\pm$ 11.88aA	23.33 $\pm$ 14.91aA	-
		LS-C	10	6.67 $\pm$ 3.85bA	5.33 $\pm$ 3.58bA	8.33 $\pm$ 4.23bA
		LS-F	9	92.78 $\pm$ 7.07aA	81.11 $\pm$ 7.64aAB	74.44 $\pm$ 27.03aB
		LS-C	10	0.0bA	0.0bA	0.0bA
	Recapture trap (CB)	LS-F	9	90.93 $\pm$ 9.02aA	80.00 $\pm$ 6.87aAB	69.26 $\pm$ 24.48aB
		LS-C	10	5.33 $\pm$ 4.50bA	14.33 $\pm$ 11.87aA	6.00 $\pm$ 3.78aA
		LS-F	9	25.19 $\pm$ 9.88aA	18.89 $\pm$ 8.33aA	7.78 $\pm$ 7.45aB
		LS-C	10	0.0bA	0.0bA	0.0aA
		LS-F	9	25.19 $\pm$ 9.88aA	18.89 $\pm$ 8.33aA	4.81 $\pm$ 5.80aB

Means followed by different letters in a column (lowercase) or in a row (uppercase) are significantly different (Tukey's HSD: $p < 0.05$)

C control, F fungus, CB cardboard trap

^aTotal mortality—dead mites (no movement) regardless the presence of fungal structures on the cadavers; Confirmed mortality—assessed using ethanol-washed mites held for fungal emergence and confirmation

Table 6 Mean ($\pm$ SD) total and confirmed mortalities (%) of *Dermanyssus gallinae* caused by *Beauveria bassiana* in cardboard (CB) traps tied to hen cages in poultry house (assessed daily for up to 6 days post-exposure)

	Trap type		N	12 h	24 h	48 h
	Capture trap (CB)					
Field trial 3		Total mortality (%) ^a	20	5.87 $\pm$ 5.02bA	4.80 $\pm$ 5.37bA	6.71 $\pm$ 7.93bA
			18	88.15 $\pm$ 12.08aAB	95.36 $\pm$ 6.27aA	84.62 $\pm$ 9.99aB
		Confirmed mortality (%) ^a	20	0.0bA	0.0bA	0.0bA
			18	83.29 $\pm$ 12.44aAB	89.28 $\pm$ 9.31aA	81.00 $\pm$ 10.94aB
	Recapture trap (CB)	Total mortality (%)	20	5.01 $\pm$ 4.57bA	5.59 $\pm$ 6.84aA	5.63 $\pm$ 5.91aA
			19	18.46 $\pm$ 12.24aA	16.30 $\pm$ 14.72aA	10.86 $\pm$ 10.87aA
Field trial 4		Confirmed mortality (%)	20	0.0bA	0.0bA	0.0bA
			19	9.69 $\pm$ 9.90aA	7.72 $\pm$ 9.57aA	9.73 $\pm$ 11.15aA
	Capture trap (CB)	Total mortality (%)	30	5.64 $\pm$ 6.33bA	4.30 $\pm$ 5.12bA	6.49 $\pm$ 7.04bA
			29	91.15 $\pm$ 8.67aA	85.86 $\pm$ 20.37aAB	77.72 $\pm$ 13.14aB
		Confirmed mortality (%)	30	0.0bA	0.0bA	0.0bA
			29	83.00 $\pm$ 8.63aA	79.98 $\pm$ 19.64aA	72.07 $\pm$ 12.81aB
	Recapture trap (CB)	Total mortality (%)	30	5.33 $\pm$ 6.27bA	4.46 $\pm$ 5.51bA	4.44 $\pm$ 5.13aA
			29	16.34 $\pm$ 13.25aAB	19.09 $\pm$ 12.45aA	11.06 $\pm$ 12.67aB
		Confirmed mortality (%)	30	0.0bA	0.0bA	0.0bA
			29	10.13 $\pm$ 10.26aA	11.79 $\pm$ 8.73aA	4.77 $\pm$ 7.28aB

Means followed by different letters in a column (lowercase) or in a row (uppercase) are significantly different (Tukey's HSD: $p < 0.05$)

C control, F fungus

^aTotal mortality—dead mites (no movement) regardless the presence of fungal structures on the cadavers; Confirmed mortality—assessed using ethanol-washed mites held for fungal emergence and confirmation

the mites after blood feeding. Traps containing *B. bassiana* conidia were not repellent, because preliminary arena choice-tests with fungal treated and untreated inert supports were equally visited by the mites (data not shown). Moreover, the mites that entered the device were successfully infected by the fungus, regardless of trap type. In spite of increasing the number of conidia adhered to cardboard traps, SPLAT wax did not induce higher levels of mite mortality than traps without wax in laboratory bioassays. Probably, the amount of conidia in CB traps without the wax was sufficient to infect and kill the individuals hidden inside the trap. On the other hand, the higher amount of conidia in LS traps and the web structure of the loofah sponge increased mite infection compared to CB traps, indicating that conidial delivery might be more efficient on this support.

Acaropathogenic fungi have been shown to successfully infect the poultry red mite exposed to conidial suspensions in field (Tavassoli et al. 2011; Steenberg and Kilpinen 2014; Kasburg et al. 2016). However, harsh environmental conditions found in poultry houses, especially heat stress, may present a major challenge for the use of mycopesticides against *D. gallinae*. For instance, the incidence of mycosis caused by *Metarhizium brunneum* on poultry red mites after surface treatment with unformulated conidia in water were significantly reduced under poultry house simulated conditions compared to optimal conditions, suggesting the need for new formulations and delivery strategies to minimize the negative impact of abiotic effects on the fungus (Tomer et al. 2018). In both our field trials, the air temperature inside the poultry house reached over 30 °C, still CB and LS traps induced high levels of mite infection. The traps seem to isolate the fungus from the heat of the metallic cage surface and provide ventilation through the inert support structure, avoiding high temperature peaks that damage conidia. Although conidial viability in both trap types decreased over time when exposed to unrefrigerated conditions in the laboratory, around 80% of the conidia in CB and LS traps were still viable after 21 and 60 days at 26 and 32 °C, respectively. Thus, the autoinoculation device might be able to infect mite populations up to at least 3 weeks in poultry houses depending on the temperature variation. In our field trials, CB and LS capture traps remained active (more than 75% of mite mortality) up to 2 and 5 days, respectively. The accumulation of dust particles, cleaning practices in the cages by the farmers and moisture conditions to help the infection process are important factors that may negatively influence the fungal efficacy after conventional sprays. It is noteworthy that all these factors can also be mitigated by the use of the autoinoculation device here proposed.

Infected mites were recaptured 24 h after fungal-treated traps were removed, indicating that part of the population had dispersed to other shelters after visiting the trap containing *B. bassiana* and revisited the same place after meal. The percentages of infected mites in recapture traps were low (between 5 and 25%) but probably represented a substantial part of the whole population, considering the great infestation found in the poultry houses during the trials. Moreover, infected individuals may die in other mite aggregations and eventually produce secondary infections. However, the potential horizontal transmission and the dynamic of this fungal disease in mite populations must still be clarified.

The efficiency of poultry red mite control depends on several environmental factors or poultry house conditions, and future improvements to mite control will likely require integrated strategies (Pritchard et al. 2015). Our results are important for development of a non-chemical alternative to incorporate into an integrated management strategy to control *D. gallinae*. The attract-and-infect device proposed here can be easily handled and periodically replaced, avoiding the animal stress caused in spray or dust applications. Further studies are needed on how the number of traps and their distribution in poultry houses

under different infestation levels may help to mitigate the problem caused by the mite in a long-term control strategy.

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References

- Abbas RZ, Colwell DD, Iqbal Z, Khan A (2014) Acaricidal drug resistance in poultry red mite (*Dermanyssus gallinae*) and approaches to its management. *World Poultry Sci J* 70:113–124
- Barimani A, Youssefi MR, Tabari MA (2016) Traps containing carvacrol, a biological approach for the control of *Dermanyssus gallinae*. *Parasitol Res* 115:3493–3498
- Beys-da-Silva WO, Rosa RL, Berger M, Coutinho-Rodrigues CJB, Vainstein MH, Schrank A, Bittencourt VREP, Santi L (2020) Updating the application of *Metarhizium anisopliae* to control cattle tick *Rhipicephalus microplus* (Acari: Ixodidae). *Exp Parasitol* 208:107812
- Chandler D, Davidson G, Pell JK, Ball BV, Shaw K, Sunderland KD (2000) Fungal biocontrol of Acari. *Biocontrol Sci Technol* 10:357–384
- Chauve C (1998) The poultry red mite *Dermanyssus gallinae* (De Geer, 1778): current situation and future prospects for control. *Vet Parasitol* 79:239–245
- Chirico J, Tauson R (2002) Traps containing acaricides for the control of *Dermanyssus gallinae*. *Vet Parasitol* 110:109–116
- Eladl AH, Hamed HR, El-Shafei RA (2017) Prevalence of mites and their impact on laying hen (*Gallus gallus domesticus*) farm facilities in Egypt, with an analysis of deltamethrin residues in eggs and tissue. *Avian Pathol* 47:161–171
- Entrekin DL, Oliver JH (1982) Aggregation of the chicken mite, *Dermanyssus gallinae* (Acari: Dermanyssidae). *J Med Entomol* 19:671–678
- Fernandes EKK, Bittencourt VREP, Roberts DW (2012) Perspectives on the potential of entomopathogenic fungi in biological control of ticks. *Exp Parasitol* 130:300–305
- Flochlay AS, Thomas E, Sparagano OA (2017) Poultry red mite (*Dermanyssus gallinae*) infestation: a broad impact parasitological disease that still remains a significant challenge for the egg-laying industry in Europe. *Parasit Vectors*. <https://doi.org/10.1186/s13071-017-2292-4>
- George DR, Finn RD, Graham KM, Mul M, Maurer V, Valiente Moro C, Sparagano AO (2015) Should the poultry red mite *Dermanyssus gallinae* be of wider concern for veterinary medical science. *Parasit Vectors*. <https://doi.org/10.1186/s13071-015-0768-7>
- Hamscher G, Prieb B, Hartung J, Nogosseck MI, Glunder G, Nau H (2003) Determination of propoxur residues in eggs by liquid chromatography-diode array detection after treatment of stocked housing facilities for the poultry red mite (*Dermanyssus gallinae*). *Anal Chim Acta* 489:19–26
- Immediato D, Camarda A, Iatta R, Puttilli MR, Ramos RAN, Di Paola G, Giangaspero A, Otranto D, Cafarchia C (2015) Laboratory evaluation of a native strain of *Beauveria bassiana* for controlling *Dermanyssus gallinae* (De Geer, 1778) (Acari: Dermanyssidae). *Vet Parasitol* 212:478–548
- Immediato D, Figueredo LA, Iatta R, Camarda A, de Luna RLN, Giangaspero A, Brandão-Filho SP, Otranto D, Cafarchia C (2016) Essential oils and *Beauveria bassiana* against *Dermanyssus gallinae* (Acari: Dermanyssidae): towards new natural acaricides. *Vet Parasitol* 229:159–165
- Immediato D, Iatta R, Camarda A, Giangaspero A, Capelli G, Figueredo LA, Otranto D, Cafarchia C (2017) Storage of *Beauveria bassiana* conidia suspension: a study exploring the potential effects on conidial viability and virulence against *Dermanyssus gallinae* De Geer, 1778 Acari: Dermanyssidae. *Ann Biol Sci* 5:69–76
- Kaoud HA (2010) Susceptibility of poultry red mites to entomopathogens. *Int J Poult Sci* 9:259–263
- Kasburg CR (2016) Selection and characterization of isolates of entomopathogenic fungi aiming to control red mite *Dermanyssus gallinae* (Acari: Dermanyssidae). Dissertation, Universidade Estadual do Oeste do Parana, Brazil
- Kasburg CR, Alves LFA, Oliveira DGP, Rohde C (2016) Activity of some Brazilian isolates of entomopathogenic fungi against the poultry red mite *Dermanyssus gallinae* De Geer (Acari: Dermanyssidae). *Braz J Poultry Sci* 18:457–460

- Kilpinen O, Roepstorff A, Permin A, Norgaard-Nielsen G, Lawson LG, Simonsen HB (2005) Influence of *Dermanyssus gallinae* and *Ascaridia galli* infections on behaviour and health of laying hens (*Gallus gallus domesticus*). Brit Poultry Sci 46:26–34
- Koenraadt CJM, Dicke M (2010) The role of volatiles in aggregation and host-seeking of the haematophagous poultry red mite *Dermanyssus gallinae* (Acari: Dermanyssidae). Exp Appl Acarol 50:191–199
- Lammers GA, Bronneberg RGG, Vernooij JCM, Stegeman JA (2017) Experimental validation of the AVIVET trap, a tool to quantitatively monitor the dynamics of *Dermanyssus gallinae* populations in laying hens. Poult Sci 96:1563–1572. <https://doi.org/10.3382/ps/pew428>
- Lundh J, Wiktelius D, Chirico J (2005) Azadirachtin-impregnated traps for the control of *Dermanyssus gallinae*. Vet Parasitol 130:337–434
- Marangi M, Morelli V, Pati S, Camarda A, Cafiero MA, Giangaspero A (2012) Acaricide residues in laying hens naturally infested by red mite *Dermanyssus gallinae*. PLoS ONE 7:e31795
- Nordenfors H, Chirico J (2001) Evaluation of a Sampling Trap for *Dermanyssus gallinae* (Acari: Dermanyssidae). J Econ Entomol 94:1617–1621
- Pritchard J, Kuster T, Sparagano O, Tomley F (2015) Understanding the biology and control of the poultry red mite *Dermanyssus gallinae*: a review. Avian Pathol 44:143–215
- Rehner SA, Minnis AM, Sung GH, Luangsa-ard JJ, Devotto L, Humber RA (2011) Phylogeny and systematics of the anamorphic, entomopathogenic genus *Beauveria*. Mycologia 103:1055–1073
- Sparagano O, Pavlicevic A, Murano T, Camarda A, Sahibi H, Kilpinen O, Mul M, van Emous R, Bouquin S, Hoel K, Cafiero MA (2009) Prevalence and key figures for the poultry red mite *Dermanyssus gallinae* infections in poultry farm systems. Exp Appl Acarol 48:3–10
- Sparagano OA, George DR, Harrington DW, Giangaspero A (2014) Significance and control of the poultry red mite, *Dermanyssus gallinae*. Annu Rev Entomol 59:447–466
- Steenberg T, Kilpinen O (2014) Synergistic interaction between the fungus *Beauveria bassiana* and desiccant dusts applied against poultry red mites (*Dermanyssus gallinae*). Exp Appl Acarol 62:511–524
- Tao S, Liu WX, Li XQ, Zhou DX, Li X, Yang YF, Yue DP, Coveney RM (2009) Organochlorine pesticide residuals in chickens and eggs at a poultry farm in Beijing, China. Environ Pollut 157:497–502
- Tavassoli M, Allymehr M, Pourseyed SH, Ownag A, Bernousi I, Mardani K, Ghorbanzadegan M, Shokrpour S (2011) Field bioassay of *Metarhizium anisopliae* strains to control de poultry red mite *Dermanyssus gallinae*. Vet Parasitol 178:374–379
- Tomer H, Blum T, Arye I, Faigenboim A, Gottlieb Y, Ment D (2018) Activity of native and commercial strains of *Metarhizium* spp. against the poultry red mite *Dermanyssus gallinae* under different environmental conditions. Vet Parasitol 262:20–25